

PaperPilot – AI Research Assistant



Intelligent Paper Management & Analysis

Problem	Impact
 Information Overload	10,000+ papers published daily on arXiv alone
 Time-Consuming Literature Review	Researchers spend 100+ hours per project just reading
 Disconnected Tools	Search, reading, analysis, writing in separate apps
 Knowledge Silos	No centralized memory of what you've read
 Underutilized AI	Existing tools lack context-aware research assistance



INTRODUCING **PAPERPILOT** - An intelligent research workspace that combines paper management with context-aware AI assistance

✓ All-in-One Platform
Search, upload, analyze, and chat - ONE app

✓ Context-Aware AI
Groq Llama 3.3 70B with workspace memory

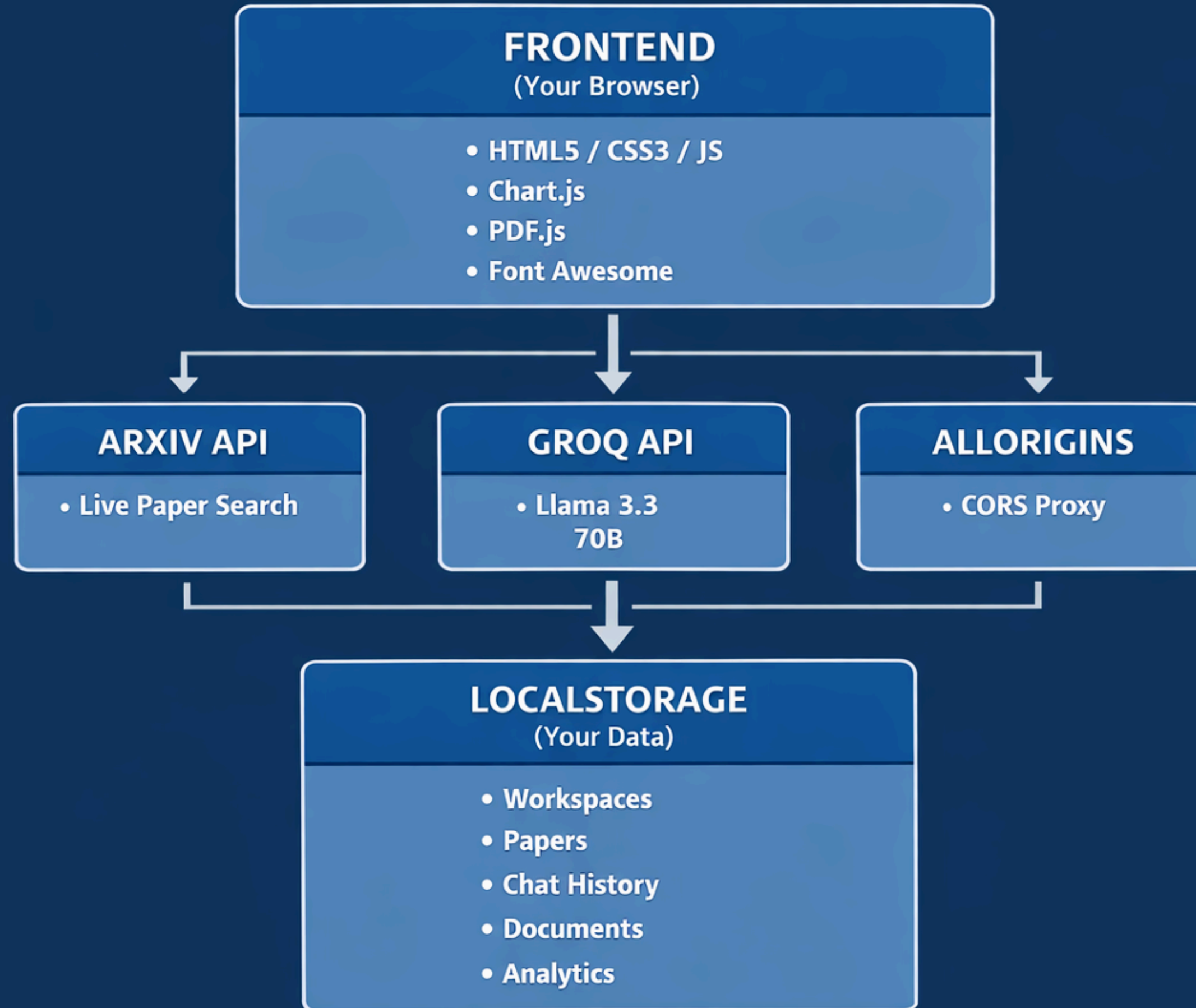
✓ Real arXiv Integration
Live search from 2.5M+ academic papers

✓ Workspace Organization
Separate contexts for different research projects

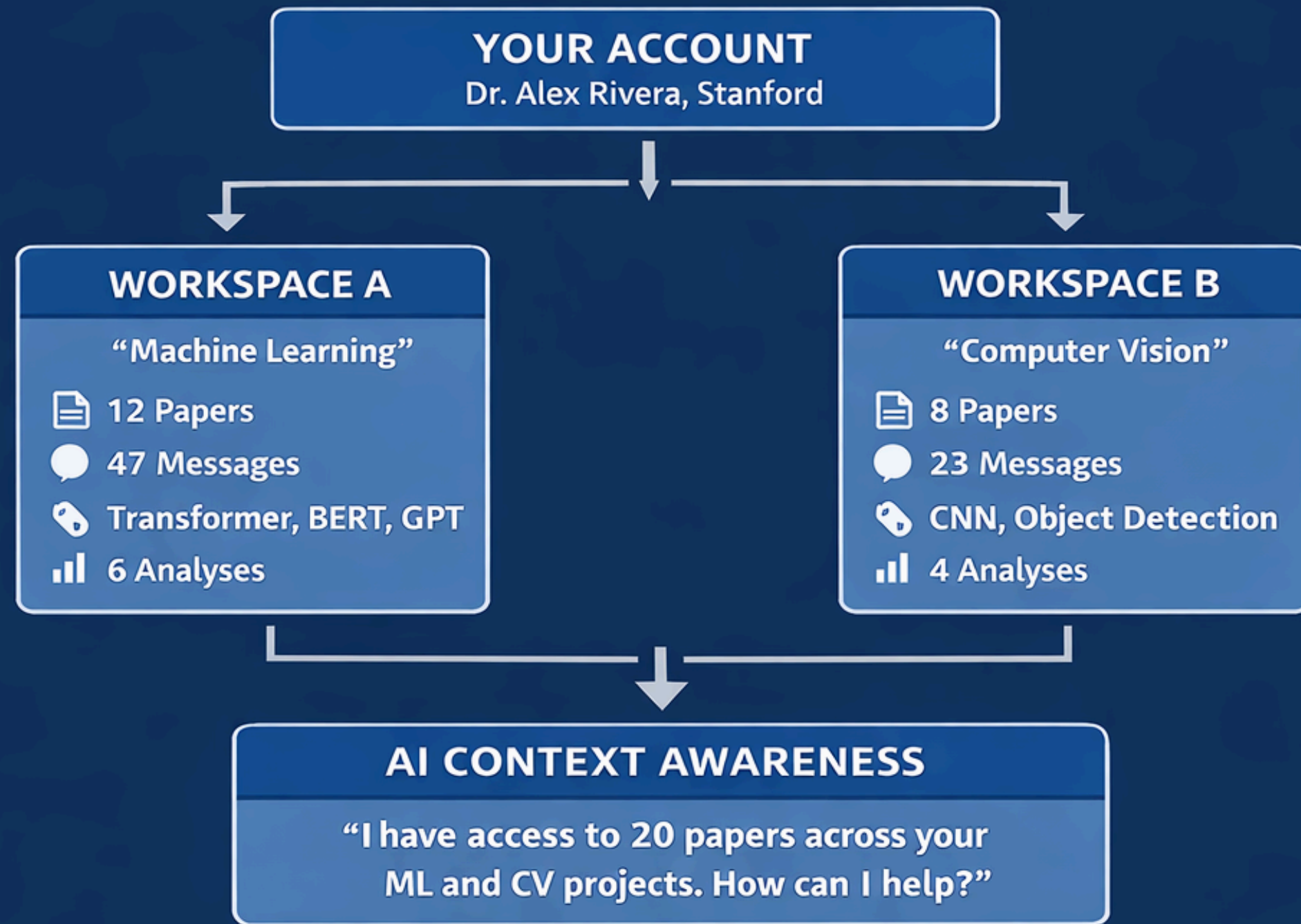
✓ Instant PDF Processing
Upload & auto-analyze in seconds

✓ Research Analytics
Track productivity and research trends

PAPERPILOT - TECHNICAL ARCHITECTURE



ISOLATED RESEARCH CONTEXTS



Each workspace has its own:

- Paper collection
- Chat history
- AI memory
- Analysis results
- No cross-contamination between projects

AI-POWERED RESEARCH TOOLS

TOOL	FUNCTION	OUTPUT
 PAPER SUMMARIZER	Condense full papers into bullet	• 3-5 key points • Methodology • Key findings
 PAPER COMPARATOR	Side-by-side methodology analysis	• Strengths • Weaknesses • Recommendations
 RESEARCH GAP ANALYZER	Identify unanswered questions	• Future research directions • Open problems

TECHNICAL STACK

CATEGORY	TECHNOLOGY	PURPOSE
FRONTEND	• HTML5 • CSS3 • Vanilla JavaScript	• Structure • Styling • Logic (No deps!)
UI/UX	• Google Fonts • Font Awesome 6	• Inter typography • 1,600+ icons
PROCESSING	• PDF.js • LocalStorage API	• PDF text extraction • Client-side storage
APIs	• arXiv.org API • Groq Cloud API	• 2.5M+ papers • LLM inference
MODEL	• arXiv.org API • Groq Cloud API	• 2.5M+ papers • LLM inference
APIs	• Llama 3.3 70B (Groq hosted)	• 70B parameters • SOTA reasoning

RESEARCH ANALYTICS & INSIGHTS

REAL-TIME METRICS



PAPERS: XX



WORKSPACES: X



ANALYSES: XX



CHATS: XXX

ACTIVITY TRENDS



Papers & Analyses over time

WORKSPACE DISTRIBUTION



Papers by Workspace

TOP RESEARCH KEYWORDS

- Transformer (12)
- Attention (10)
- BERT (8)
- Fine-tuning (7)
- Pre-training (6)

PRODUCTIVITY SUMMARY



Reading time saved: XXh



Analysis time saved: XXh



Papers per week: X.X



Active streak: X days



Thank You